

1. INPUT SPECIFICATIONS

Required capacity	100 t/h
Lift height	25 m
Material	Clinker
Bulk density	1500 kg/m3
Service environment	Dry
Site wind pressure	800 Pa

2. PERFORMANCE SUMMARY

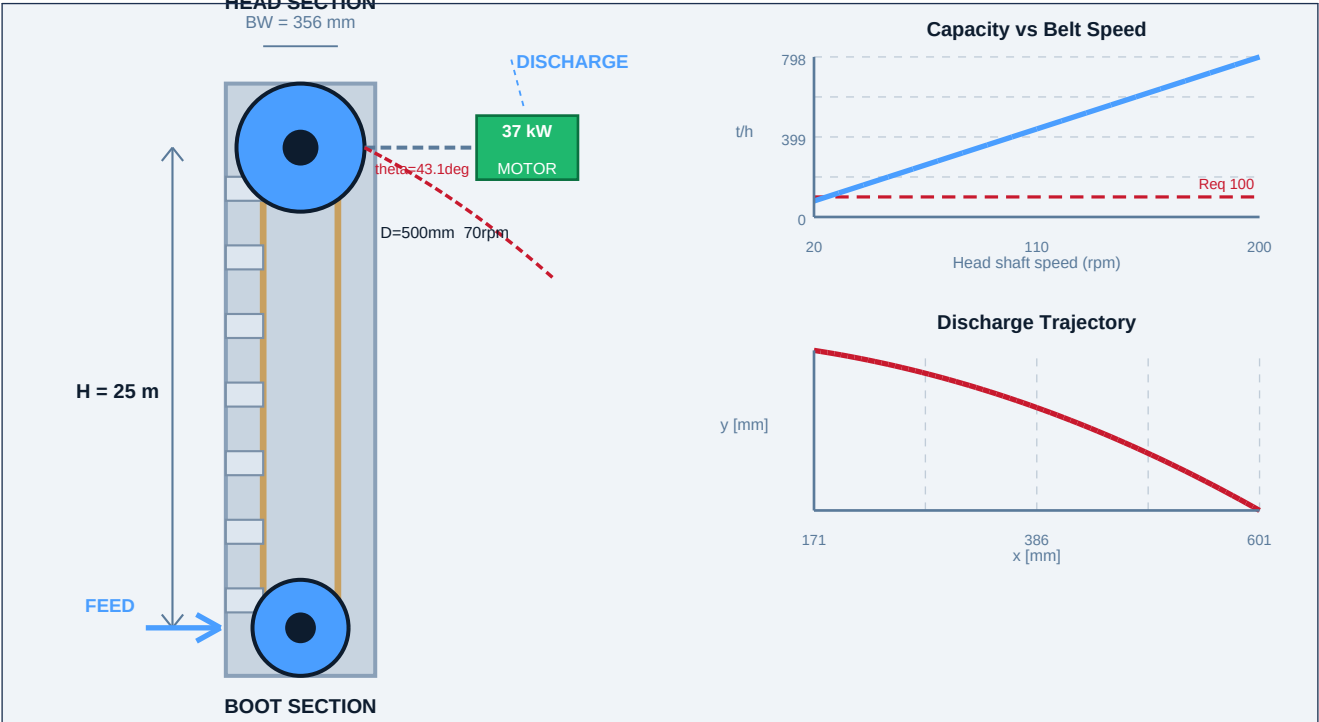
Capacity 279.3 t/h	Belt Speed 1.83 m/s	Total Power 25.0 kW	Motor Selected 37 kW	Centrifugal Ratio 1.369 —	Bearing L10 180,226 h
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3. PROCESS LEVEL OUTPUTS

Design Configuration

Head pulley dia.	500 mm	Belt type	EP
Head shaft speed	70 rpm	Belt friction mu	—
Boot pulley dia.	— mm	Wrap angle	— deg
Fill factor	75%	Service factor	—

System Schematic & Performance Charts



Achieved capacity	279.3 t/h	Discharge angle	43.1 deg from vertical
Belt speed	1.83 m/s	Belt width	356 mm

Bucket spacing	0.228 m	Recommended fill	75.0%
Centrifugal ratio	1.369	Stream spread	0.124

4. POWER BREAKDOWN

Boot equivalent height (D_boot x Leq)	2.40 m
Total equivalent height (H + H_equiv)	27.40 m
Lift power (material only)	19.03 kW
Boot scooping power (boot loading)	1.83 kW
Shaft power (total load at head shaft)	20.85 kW
Drive losses (gearbox + bearings + belt flex)	4.17 kW
Total motor power (shaft x Ceff)	25.02 kW
Design power (x service factor 1.25)	31.28 kW
Selected motor	37 kW
Gearbox ratio (at 1450 rpm input)	20.7 : 1
Drive efficiency factor Ceff	1.200
CEMA LEQ factor	8.0

5. COMPONENT DESIGN OUTPUTS

5a. Head Shaft & Drive

Material grade	ASTM A36 Mild Steel	Hub connection	Keyed (ASME B17.1)
Shaft torque	3413.9 Nm	Hub OD	162.6 mm
Section	Solid	Hub length	135.5 mm
Min shaft dia.	108.4 mm	Key b x h	25 x 14 mm
Governed by	stress	Key shear	18.59 / 42.0 MPa
Allowable shear	42 MPa (keyed)	Key bearing	66.41 / 215.0 MPa
Bearing span	686 mm	Key result	PASS
Drive arm A	309 mm	Bearing L10	180226 h
Tail arm B	377 mm		

5b. Belt & Bucket Selection

Bucket series	AC_12x8	Belt width	356 mm
Style	CC	Belt class	2 PLY
Width x depth	305 x 216 mm	Belt length (total)	52.8 m (incl. splice allowance)
Projection	203 mm	Bucket spacing (actual)	0.228 m
Volume (struck)	8.58 L	Bucket count	225 off
Bucket mass	11.00 kg	Fill factor	75%
		Material DB fill	75.0% advisory

5c. Pulley Design

Lagging type	ceramic embedded rubber	Shell min t (CEMA)	11.0 mm
Lagging thickness	16 mm	Shell min t (press)	0.6 mm
mu dry / wet	0.5 / 0.45	Shell governing t	11.0 mm (CEMA minimum)
mu operating	0.5	Shell specified t	11.0 mm (= calculated minimum, no override)
Euler limit	4.832	End disc min t	22.0 mm

Belt ratio R/T3	2.429	End disc specify	26 mm (+20%)
Slip check	PASS	Disc governed by	plate_bending
		Disc sigma bend	80.0 MPa
		Force per disc	19.48 kN
		Shaft critical speed	2171 rpm
		Operating ratio	70 rpm (3% of critical)

5d. Take-Up Design

Type	Gravity (primary)	Alt type	Screw (alternative)
Counterweight net	3270.0 kg	Screw load	37.74 kN
Counterweight gross	3597.0 kg	Min core dia.	28.3 mm
Travel required	461 mm	Turns required	77.0
— thermal	48 mm	Buckling SF	4.57
— elongation	113 mm	Buckling check	PASS

5e. Casing & Structural

Plate thickness	8.2 mm	Panel delta actual	2.99 mm
Max stiffener pitch	2065.0 mm	Panel delta allowed	4.88 mm
Recommended pitch	1755.0 mm	Panel sigma max	11.1 MPa
Deflection limit	L / 360	Panel check	PASS
Wind pressure	800.0 Pa	Bolt fatigue Goodman	0.004
		Assembly fasteners	M12 x 793 (panel-to-panel + stiffener)

5f. Discharge Chute Design

Back-plate angle	60.2 deg	Spout width	405.0 mm
Min angle (wall fric.)	40.0 deg	Throw distance	0.144 m
Angle adequate	YES	Throat velocity	0.912 m/s
Flow regime	MASS_FLOW	Geom plugging risk	LOW
Mass-flow threshold	31.5 deg	Hood radius	0.312 m
Governed by	angle of repose	Capture efficiency	0.83
Wear index	16.79 (HIGH)	Plugging risk	HIGH
Liner material	AR500 Wear Plate	Plugging index	1.205
Liner thickness	12 mm	Dust risk	MEDIUM
Liner grade	500 BHN	Recommended sensors	liner_wear_indicator , flow_sensor_acoustic, chute_blockage_detector

Chute Recommendations

* Plugging index 1.21 (HIGH) — Air cannons or fluidising pads; steep chute angle required.

* Wear index 16.8 (HIGH) — specify AR500 Wear Plate (500 BHN), 12 mm thick.

5g. Feed Design (Boot Section)

Loading method	Centrifugal – Digging / Scooping	Material depth	152.0 mm (0.75xP)
Volumetric flow	186.2 m ³ /h	Dig zone arc	236.0 mm
Inlet velocity	0.3 m/s	Dig volume	12.8 L
Inlet area req.	1724.1 cm ²	Surge volume	155.2 L (3.0s)
Inlet width	356.0 mm	Boot casing h	619.0 mm
Inlet height	152.0 mm	Floor clearance	50.0 mm

6. ENGINEERING VERIFICATION

Check	Actual	Limit	SF	Result
Capacity	279.3 t/h	100.0 t/h	2.79	✓ OK
Belt speed	1.83 m/s	0.76 – 2.03 m/s	2.41	✓ OK
Centrifugal ratio	1.369	1.00 – 2.50	–	✓ OK
Belt slip T3 / T3_min	16.04 kN	11.39 kN min	1.41	✓ OK
Headshaft radial load	38.95 kN	80 kN	2.05	✓ OK
Shaft diameter (stress)	108.4 mm calc	108.4 mm governing	1.00	i INFO
Bearing L10 life	180,226 h	20,000 h min	9.01	✓ OK
Keyway shear stress	18.59 MPa	42.0 MPa	2.26	✓ OK
Keyway bearing stress	66.41 MPa	215.0 MPa	3.24	✓ OK
Bolt fatigue Goodman ratio	0.004	< 1.000 (infinite life)	250.00	✓ OK
Lagging belt slip ratio	2.429	4.832 (e ^{mu.theta})	1.99	✓ OK
Casing panel deflection	2.99 mm	4.88 mm (L/360)	1.63	✓ OK
Casing clearance (stream vs wall)	601.4 mm stream reach	228.0 mm wall	-373.4 mm margin	✗ FAIL
Stream → chute interception	39.4 deg impact angle	2.09 m/s impact velocity	–	✓ OK

Corrective Actions

■ PLUGGING — Chute plugging probability HIGH despite adequate angle — cohesive material

1. Specify AR500 Wear Plate liner to reduce surface adhesion
2. Install vibration motor or air cannon on chute back-plate
3. Consider chute geometry with steeper secondary slope

7. DESIGN NOTES

Capacity & Selection: The elevator is configured with Series AC_12x8 buckets and achieves a throughput of 279.3 t/h against the 100 t/h requirement, providing a capacity margin of 179%. Belt speed is 1.83 m/s at the specified head shaft speed, which is within the acceptable range for the selected bucket geometry. The centrifugal discharge ratio of 1.369 is within the optimal range for clean centrifugal discharge.

Head Shaft & Hub: The head shaft minimum diameter is 108 mm, governed by stress. The shaft spans 686.0 mm between bearing centrelines. The hub outer diameter is 162.6 mm with a hub engagement length of 135.5 mm. The keyway stress check passes at the selected shaft and hub dimensions.

Pulley Lagging: Ceramic embedded rubber lagging, 16 mm thick, is specified for the head pulley. This provides a belt-to-pulley traction coefficient of 0.5 under dry service conditions. Belt slip analysis confirms the lagging selection is adequate at operating tensions.

Pulley End Disc: Pulley end discs require a structural minimum thickness of 22.0 mm (governed by plate_bending). A specified thickness of 26 mm is recommended in fabrication drawings to provide adequate construction tolerance and weld preparation allowance. Full finite element analysis or detailed annular plate calculation is recommended prior to fabrication.

Take-Up System: The gravity take-up counterweight gross mass is 3597.0 kg. The take-up frame must accommodate a minimum carriage travel of 461 mm, combining thermal expansion, belt elastic elongation, and installation clearance components. A 20% allowance above this minimum is recommended for field adjustment headroom. As an alternative for short elevators, a screw take-up requires a minimum screw core diameter of 28.3 mm. The screw buckling safety factor is adequate.

Casing & Structural: The casing plate thickness of 8.2 mm is determined by material bulk density and elevator height. Structural stiffeners at 1755.0 mm centres limit panel deflection to within acceptable serviceability limits under the 800.0 Pa design wind load. Panel deflection analysis confirms the stiffener layout is adequate.

Power & Motor: The calculated shaft power is 20.854 kW. Total motor input power including drive losses is 25.0 kW. A 37 kW motor has been selected, representing a 48% power margin above the design requirement at the specified service factor. The gearbox reduction ratio should be confirmed with the selected motor supplier based on motor synchronous speed.

Bearing Life: Head shaft bearing life is calculated at 180,226 hours, which exceeds standard requirements. Bearing selection should be confirmed from the manufacturer catalogue using the equivalent dynamic load at the specified shaft speed. Lubrication interval and replacement schedule should be established based on the supplier's rated life data for the selected bearing series.

Bucket Bolt Fatigue: Bucket mounting bolt fatigue has been assessed using the Goodman diagram method. The fatigue utilization ratio is 0.004, which is well within the infinite life threshold — bolt fatigue is not a design concern at the specified operating conditions.

Discharge Chute: The discharge chute back-plate angle of 60.2 degrees is set by the material trajectory. Flow analysis indicates mass flow conditions at the specified belt speed. AR500 Wear Plate liner, 12 mm thick, is specified based on the calculated wear index. Plugging probability is high and dust generation risk is medium for this material. The material throw distance is 0.144 m from the pulley centreline, which should be verified against the physical head section geometry.

Feed Design (Boot Section): The elevator uses centrifugal digging to fill buckets at the boot. The material depth in the boot pit should be maintained at approximately 152.0 mm (0.75 times the bucket projection) for consistent scooping. The active digging zone spans 236.0 mm of arc at the boot pulley, with an estimated digging volume of 12.8 litres. The boot surge volume is 155.2 L (3.0-second buffer). The boot casing floor should be a minimum of 619.0 mm below the boot pulley centreline to provide adequate clearance.

8. ENGINEERING SIGN-OFF

This report has been prepared in accordance with CEMA No. 375-2017 and ANSI/CEMA 550-2020. The signatory confirms that the design calculations have been reviewed and are accepted as the basis for procurement and fabrication.

DESIGNED BY Name: Designation / Company: Date: Signature STAMP / SEAL 	REVIEWED BY Name: Designation / Company: Date: Signature STAMP / SEAL 	APPROVED BY Name: Designation / Company: Date: Signature STAMP / SEAL
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Standards: CEMA No. 375-2017 · ANSI/CEMA 550-2020 · ISO 281 · ASME B17.1 · AS 4024 (where applicable)